

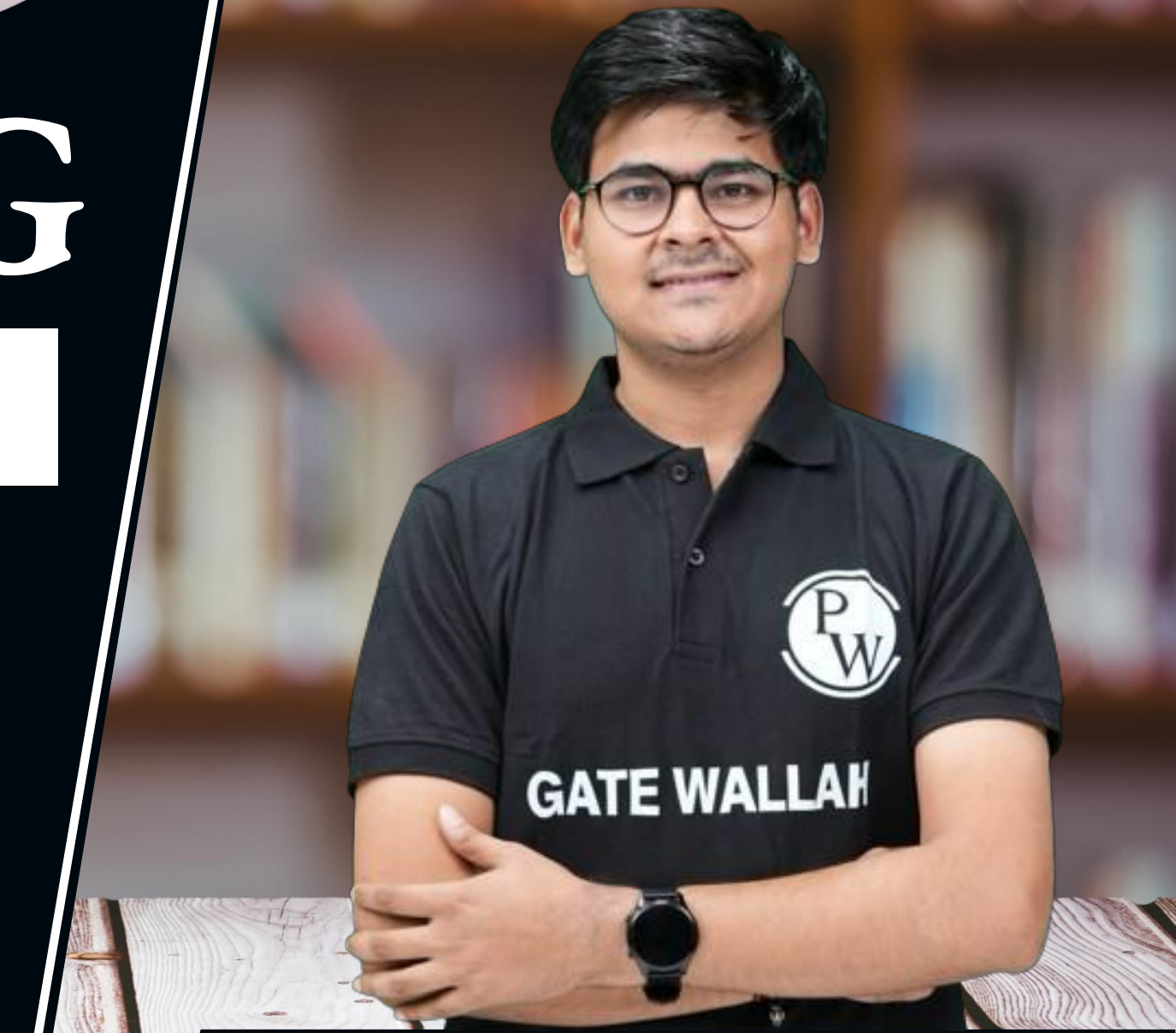
CS & IT ENGINEERING



Digital Logic

Boolean Theorems and Logical Gates

Lecture No. **2**



By- Aditya sir

Recap of Previous Lecture



Topic

→ Intro to Course
→ Intro to Boolean
Theorems



Topics to be Covered



Topic

1) AND operator

2) Properties

3) SOP & POS

forms.





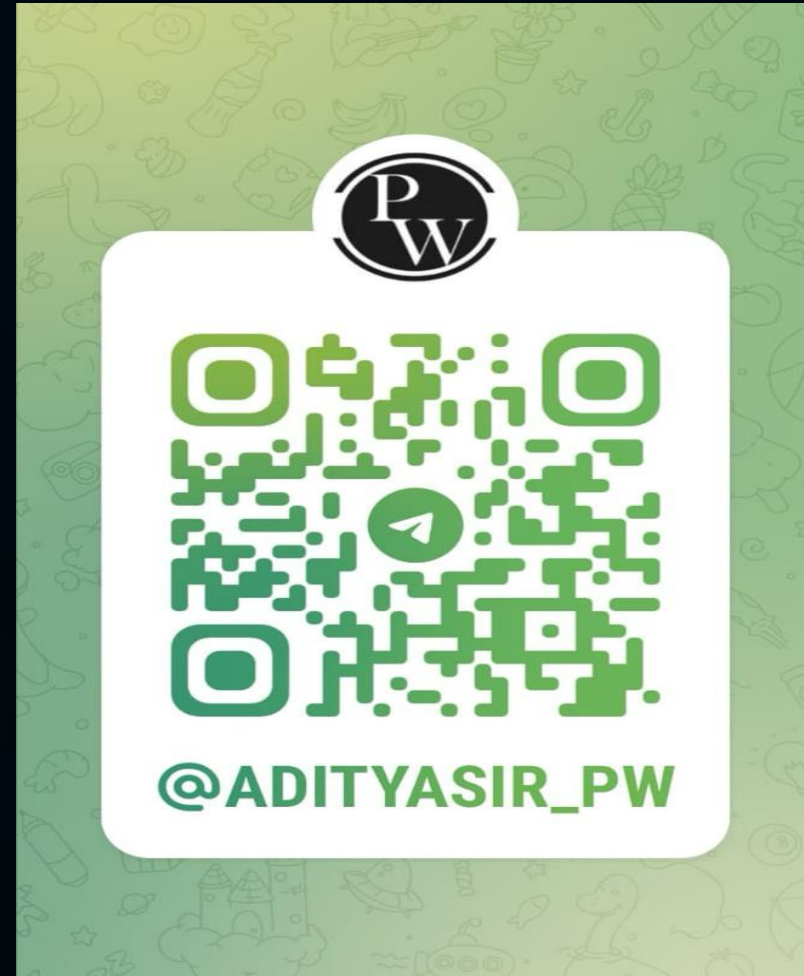
About Aditya Jain sir



1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



Telegram



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

★) AND operator: $(\cdot)$



$$1) A \cdot 0 = 0 \Rightarrow \underline{0 \cdot A = 0}$$

$$2) 0 \cdot \text{anything} = 0$$

$$3) 0 \cdot \underline{(PQ + \bar{Q}R)} = 0$$

$$4) A \cdot 1 = A$$

$$5) (\bar{A}) \cdot 1 = \bar{A} \Rightarrow \bar{A} = x \Rightarrow x \cdot 1 = x = \bar{A}$$

$$6) A \cdot A = A$$

$$7) (A \cdot A) \cdot A = (A \cdot A) = A$$

$$8) \overline{PQ} \cdot \overline{PQ} = \overline{PQ}$$

$$9) A \cdot \overline{A} = 0 \quad \rightarrow \quad PQ \cdot \overline{PQ} = 0$$

$\downarrow$ $\downarrow$
 1 0 x $\overline{x}$

$1 \cdot 0 = 0$ $x \cdot \overline{x} = 0$

$$\begin{array}{l} 0 \cdot 0 = 0 \\ 0 \cdot 1 = 0 \\ 1 \cdot 0 = 0 \end{array}$$

$$1 \cdot 1 = 1$$

Imp Properties:

1) 'AND' is distributive over 'OR'

$$P \cdot (Q + R) = P \cdot Q + P \cdot R$$

Validate

Truth table

all combinations will
give same result

0 0	→ 0	→ 0
0 1	→ 0	→ 0
1 0	→ 1	→ 1

2) 'OR' is distributive over 'AND'

$$\underline{P + (Q \cdot R) = (P + Q) \cdot (P + R)}$$

eg: $P=1, Q=0, R=0$

$$\rightarrow P + (Q \cdot R)$$

$$= 1 + 0$$

$$= 1$$

$$(P + Q) \cdot (P + R)$$

$$= 1 \cdot 1$$

$$= 1$$

use-case

$$P + (QR) = (P+Q) \cdot (P+R)$$

ex 1 :-

$$\underline{\underline{P + (\bar{P}Q)}} = \underline{\underline{(P + \bar{P}) \cdot (P + Q)}}$$

$$= 1 \cdot (P + Q)$$

$$\Rightarrow \underline{\underline{P + Q}}$$

$$A + (BC) = \underline{\underline{(A+B)}} \underline{\underline{(A+C)}} = A + (BC)$$

$$3) \quad \overline{P} + (Q \cdot P)$$

$$= (\overline{P} + Q) \cdot \underline{(\overline{P} + P)}$$

$$= (\overline{P} + Q) \cdot 1$$

$$= (\overline{P} + Q)$$

$$4) \quad (A+B) \cdot (\underline{A+C}) \cdot (\underline{A+D})$$

$$= (A \cdot A + A \cdot C + B \cdot A + B \cdot C) (A+D)$$

mentos Zindgi 😊

$$A + (B \cdot C \cdot D)$$

Practice Questions:

$$1) (P + \overline{Q}) \cdot (\overline{P} + R) \cdot (\overline{P} + \overline{Q})$$

$$= (P + \overline{Q}) (\overline{P} + \overline{Q}) (\overline{P} + R)$$

$$= (\overline{Q} + \underline{P \cdot \overline{P}}) (\overline{P} + R)$$

$$= \overline{Q} \cdot (\overline{P} + R) = \overline{P} \overline{Q} + R \overline{Q}$$

$$2) (\bar{P} + \bar{Q}R) \cdot (P + \bar{Q})$$

$$A \cdot B \cdot \bar{A} = AB$$

$$A \cdot AB$$

$$AB$$

$$= \bar{P} \cdot P + \bar{P} \bar{Q} + \bar{Q} R P + \bar{Q} R \bar{Q}$$

$$= 0 + \bar{P} \bar{Q} + \bar{Q} R P + \bar{Q} R$$

$$= \bar{P} \bar{Q} + \bar{Q} R (P + 1)$$

$$= \bar{P} \bar{Q} + \bar{Q} R$$

$$= \bar{Q} (\bar{P} + R)$$

$$3) \quad (\underline{A+B+C}) \cdot (\underline{A+B+\bar{C}}) \cdot (\underline{\bar{A}+B+C})$$

$$= \left[\underline{(A+B)} + \underline{(C \cdot \bar{C})} \right] \cdot (\bar{A} + B + C)$$

$$= \underline{(A+B)} \cdot (\bar{A} + B + C)$$

$$= B + (A \cdot (\bar{A} + C))$$

$$= B + \underline{A \cdot \bar{A}} + AC$$

$$\Rightarrow \boxed{B + AC}$$

$$4) \quad (\bar{P} + \bar{Q} + R) (P + Q + \underline{R}) (\underline{P + \bar{Q} + R})$$

$$= (\bar{P} + \bar{Q} + R) (\underline{P + \bar{Q} + R}) \cdot (P + Q + R)$$

$$= \left[(\bar{Q} + R) + \underbrace{(P \cdot \bar{P})}_0 \right] (P + Q + R)$$

$$= (\bar{Q} + R) (P + Q + R)$$

$$= R + (\bar{Q} \cdot (P + Q))$$

$$= R + (\bar{Q}P + \bar{Q}Q)$$

$$= \underline{\underline{P\bar{Q} + R}} \quad \checkmark$$

App 2 :-

$$(\bar{P} + \bar{Q} + R) \overbrace{(P + Q + R)}^{\text{Redundant}} \overbrace{(P + \bar{Q} + R)}^{\text{Redundant}}$$

$$= (\bar{P} + \bar{Q} + R) \left[\underline{(P + R + Q)} \underline{(P + R + \bar{Q})} \right]$$

$$= (\bar{P} + \bar{Q} + R) \left[(P + R) + \underbrace{(Q \cdot \bar{Q})}_0 \right]$$

$$= (\bar{P} + \bar{Q} + R) \underline{(P + R)}$$

$$= R + ((\bar{P} + \bar{Q}) \cdot P)$$

$$= R + \underbrace{(P\bar{P} + P\bar{Q})}_0 = R + P\bar{Q} = \underline{\underline{P\bar{Q} + R}}$$

★ DeMorgan Theorem : $\left\{ \begin{array}{l} + \longrightarrow \cdot \\ \cdot \longrightarrow + \end{array} \right\}$

$$\rightarrow 1) \quad \overline{P+Q+R} = \overline{P} \cdot \overline{Q} \cdot \overline{R}$$

$$\begin{aligned} \rightarrow 2) \quad \overline{PQ+R} &= \overline{P \cdot Q} \cdot \overline{R} \\ &= (\overline{P} + \overline{Q}) \cdot \overline{R} \end{aligned}$$

$$3) \quad \overline{P \cdot Q \cdot R} = \overline{P} + \overline{Q} + \overline{R}$$

$$4) \quad \overline{(P+Q) \cdot R} = \overline{P+Q} + \overline{R}$$

$$= \overline{P} \cdot \overline{Q} + \overline{R}$$

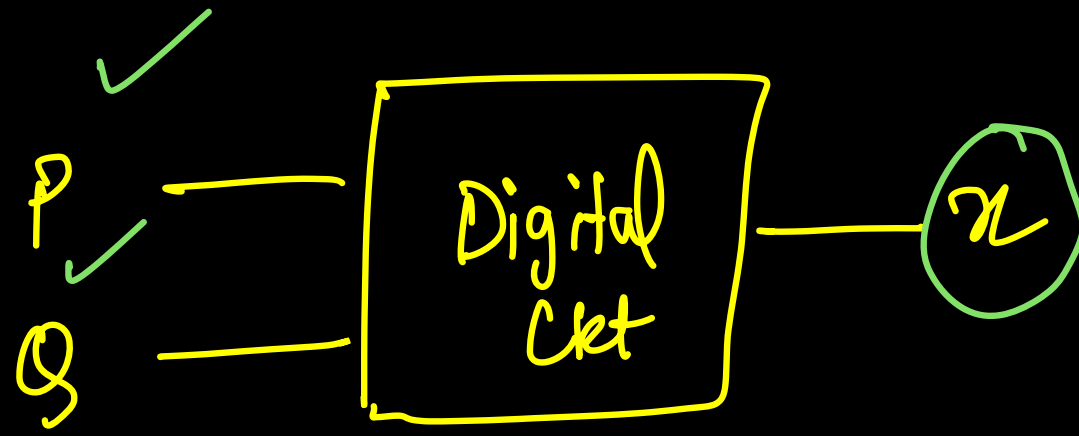
$$5) \quad \overline{P \cdot S \cdot (P \cdot Q + R)} = \overline{P} + \overline{S} + \overline{(P \cdot Q + R)}$$

$$= \overline{P} + \overline{S} + (\overline{P \cdot Q} \cdot \overline{R})$$

$$= \overline{P} + \overline{S} + ((\overline{P} + \overline{Q}) \cdot \overline{R})$$

$$= \overline{P} + \overline{S} + \overline{R} \cdot \overline{P} + \overline{R} \cdot \overline{Q}$$

* SOP & POS



1) Sum of Product terms (SOP)

1
term

$$0 \rightarrow \bar{P}$$

$$1 \rightarrow P$$

Σ

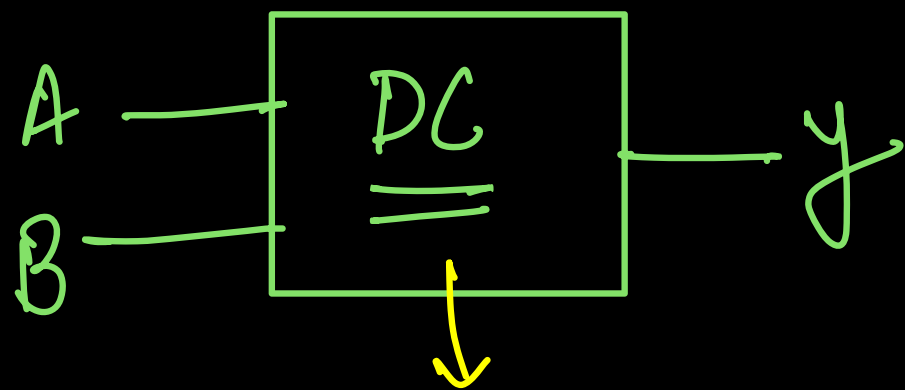
Representation

2) Product of Sum terms (POS)

0
term

$$0 \rightarrow P$$

$$1 \rightarrow \bar{P}$$



(known) given

$F(P, Q)$

SOP

POS

A

B

y

$\bar{A} \cdot \bar{B}$

$\rightarrow$

$A + B$

0

0

(0)

0

$\bar{A} \cdot B$

$\rightarrow$

$A + \bar{B}$

0

1

(1)

1

$A \cdot \bar{B}$

$\rightarrow$

$\bar{A} + B$

1

0

(2)

0

$A \cdot B$

$\rightarrow$

$\bar{A} + \bar{B}$

1

1

(3)

0

AB

AB	00	0
	01	0
	10	0
	11	1

SOP:

$$y = \sum \underline{(1)} = \underline{\bar{A}B}$$

01
0 → 3

POS
repr

$$y = \prod (0, 2, 3)$$
$$= (A + \underline{B}) \cdot (\bar{A} + \underline{B}) \cdot (\bar{A} + \underline{B})$$

$$= (B + \underline{A\bar{A}})(\bar{A} + \underline{\bar{B}})$$
$$= B \cdot (\bar{A} + \underline{\bar{B}}) = \underline{\bar{A}B}$$

Practice :-

$\underbrace{0 \rightarrow 3}_{\text{SOP}}$

POS

$\overline{P} \overline{Q}$
1 1

$$y(P, Q) = \sum (0, 1, 2) = \prod (3)$$

$$(\overline{P} + \overline{Q})$$

$$(\overline{P} \cdot \overline{Q} + \overline{P} Q + P \overline{Q})$$

$$= \overline{P} (\overline{Q} + Q) + P \overline{Q}$$

$$= \overline{P} + P \overline{Q}$$

$$= (\overline{P} + P) \cdot (\overline{P} + \overline{Q}) = \overline{P} + \overline{Q}$$

$$3) \quad y(P, Q) = \sum_{\text{SOP}} (1, 2) = \prod_{\text{POS}} (0, 3)$$

$\swarrow \quad \searrow$
 $01 \quad 10$

 $\swarrow \quad \searrow$
 $00 \quad 11$

$$= (\bar{P} \cdot Q + P \bar{Q})$$

$$(P + Q) \cdot (\bar{P} + \bar{Q})$$

$$= \underbrace{P \cdot \bar{P}} + P \cdot \bar{Q} + \bar{P} \cdot Q + \underbrace{Q \cdot \bar{Q}}$$

$$\rightarrow P \bar{Q} + \bar{P} Q$$

★ Redundancy Theorem :-

1) $\rightarrow \{ \underline{AB} + \underline{\bar{A}C} + \underline{BC} \}$

$\rightarrow$ Redundant term

$= AB + \bar{A}C$

2) $A \underline{B} + \underline{\bar{A}C} + \underline{\bar{B}C}$

A ✓
B ✓
C ✗

$$3) \quad \underbrace{AB + \bar{A}\bar{C}}_{\text{Redundant (ignore)}} + \underline{B\bar{C}} = \underline{AB + \bar{A}\bar{C}}$$

$$4) \quad \underbrace{\bar{A}B + \bar{A}\bar{C}}_{\text{redundant}} + B\bar{C} = \underline{\bar{A}\bar{C} + B\bar{C}}$$

$A \times$
 $B \times$
 $C \checkmark$

5)

$$\overline{A}\overline{C} + \overline{B}\overline{C} + A\overline{B}$$

$$= \overline{A}\overline{C} + A\overline{B}$$

A ✓

B

C

redundant

H.W $\longrightarrow$ Simplify

1) $\bar{A}B + AC + \bar{B}C$

A) $\bar{A}B + AC$

B) $\bar{A}B + C$

C) $AC + \bar{B}C$

D) $\bar{A}B + \bar{B}C$

2)

$$A\bar{B} (\bar{A}B + \bar{B}C + A\bar{B}D + A\bar{B}\bar{D})$$

A) $(\bar{A} + B)$

B) $(\bar{A} + B)(B + \bar{C})$

C) $(\bar{A} + B)(A + \bar{B})$

D) $\bar{A}B + \bar{B}D$

$$3) \quad A + BC + \bar{A}C$$

$$A) \quad (A + C)$$

$$B) \quad (A + B)(B + C)$$

$$C) \quad (A + B)(\bar{A} + C)$$

$$D) \quad C(\bar{A} + B)$$

$$4) \quad \overline{(A+B)} \cdot (\overline{B+C}) = ?$$

A) $\overline{B}(A+C)$

B) $C(A+B)$

C) $B(A+C)$

D) $A(B+C)$

$$5) \quad \overline{A}\overline{B} + AC + \overline{B}C = ?$$

$$A) \quad AC + \overline{B}C$$

$$B) \quad \overline{A}\overline{B} + \overline{B}C$$

$$C) \quad \overline{A}\overline{B} + AC$$

$$D) \quad AB + C$$



2 mins Summary



Topic

DeMorgan Thm

Topic

Redundancy Thm

Topic

SOP

Topic

POS

Topic

Practice Questions



THANK - YOU